AI-Powered Full Stack Developer Program

CLASSWORK BOOKLET

Week 2 · Day 2

Semantic HTML, Accessibility, Media → CSS Basics

Every classwork task in order: D2-00 … then fast finishers FF1–FF6 (no solutions)

Momo House · TSI Canteen · Aarav · Sunita · Didi · Rajan

Stage 1, AI as Tutor: AI may explain; you type every line.

Tech Skills Institute (TSI) Nepal · Kathmandu

D2-00 · Outfit swap

File: classwork/D2-00-outfit-swap/README.md

Time: 3 min · **Type:** Modify

Goal: See that one HTML page can wear many outfits. You change one word and the whole page changes.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

- ☐ Find line 7 in `index.html`: `<link rel="stylesheet" href="css/outfit-traditional.css">`.
- ☐ Change `outfit-traditional.css` to `outfit-modern.css`. Save (Ctrl+S). Look at the page.
- ☐ Change it to `outfit-night.css`. Save. Look again.
- ☐ Answer in your head: did you change any text inside `<body>`?

Skeleton = HTML (what is on the page). Outfit = CSS (how it looks). Today you learn to sew the outfit.

Done looks like

You saw three different looks, and you never touched the `<body>`. **Same skeleton, new outfit.**

Stuck?

1. Nothing changes? Did you save? Is Live Server running (look for **Port: 5500** at the bottom of VS Code)?
2. The page is plain black and white? Check the spelling of the file name. Look in the `css/` folder: the names must match exactly.
3. Still stuck? Copy the name from the file list: right-click the CSS file → **Copy Relative Path**, then paste it after `href="` and delete the part before `css/`.

Fast?

Open the three CSS files side by side. Find one selector that is in all three. What is different in each one?

D2-01 · Div soup → semantic

File: classwork/D2-01-div-soup-to-semantic/README.md

Time: 7 min · **Type:** Modify

Goal: Turn 7 unlabeled boxes (`<div class="…">`) into boxes with real names.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

- ☐ `<div class="header">` → `<header>`
- ☐ `<div class="nav">` → `<nav>`
- ☐ `<div class="main">` → `<main>`
- ☐ `<div class="section" id="today">` → `<section id="today">`
- ☐ `<div class="article" id="review">` → `<article id="review">`
- ☐ `<div class="aside" id="hours">` → `<aside id="hours">`
- ☐ `<div class="footer">` → `<footer>`
- ☐ Delete each `class="…"` you no longer need. **Keep the `id`s**: the nav links jump to them.
- ☐ Open F12 → **Elements**. You now see `header`, `nav`, `main` … instead of 7 × `div`.

Rooms with names. A house plan that says "room 1, room 2, room 3" tells you nothing. "Kitchen, bedroom, bathroom" tells you everything. `<div>` is "room 1". `<header>` is "the front door".

Done looks like

The page looks exactly the same as before. The Elements panel shows 7 named boxes and no `div` at all.

Stuck?

1. **Linked editing:** click inside the word `div` of the opening tag and type the new name. The closing `</div>` changes by itself. (It is on in this folder's `.vscode/settings.json`.)
2. Not sure which `</div>` belongs to which box? Click on the opening tag: VS Code highlights its partner.
3. The page still works if a `class` is left in, but tidy code has no labels it does not need.

Fast?

Answer: who reads these names? (Google, screen readers, AI tools, your teammates.) Then start D2-02.

D2-02 · Semantic misuse

File: classwork/D2-02-semantic-misuse-bughunt/README.md

Time: 8 min · **Type:** 🐞 Bug-hunt

Goal: Learn when **not** to use each semantic tag.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

1. Read the page in the browser, then read the code.
2. For every semantic tag, ask: "Is this the right name for this box?"
3. Mend each mistake in `index.html`. Save and check the page again.

The bugs

There are 5 bugs. Find at least 3.

Hints, one per bug type:

- **Count the rooms:** some rooms must exist only once on a page.
- **Signposts:** `<nav>` is for links that take you somewhere. Is that true here?
- **Labels:** a `<section>` needs a heading that names it.
- **Main vs side:** what is the most important content? Is it in the right box?
- **Real controls:** press the **Tab** key again and again. Does focus reach every button?

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

One `<main>`, a `<nav>` only around links (or no nav), every section has a heading, the blog post is the main content, and Tab reaches a real Subscribe button.

Scoreboard

```
I found __ / 5. The reveal taught me: ________
```

Stuck?

1. Open F12 → Elements and look at the boxes in order. Which box holds what?
2. Read the "When NOT to use each tag" slide or `notes/01-semantic-html5.md`.
3. Ask the AI tutor: "When should I use aside instead of main?" Then mend it yourself.

Fast?

Try **D2-FF5** Lost & Found (it has more semantic and a11y mistakes).

D2-03 · A11y audit: TSI Admissions

File: classwork/D2-03-a11y-audit/README.md

Time: 10 min · **Type:** ⚽ Bug-hunt

Goal: Make the TSI Admissions page work for everyone, including people who use a screen reader.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

1. **Tab walk:** click in the address bar, then press **Tab** again and again. Where does focus go? Can you tell where each link goes?
2. **Lighthouse:** F12 → **Lighthouse** tab → tick **Accessibility** only → **Analyze page load**. Read the red items.
3. **Read the code** for the problems no tool can see.
4. Mend each problem in `index.html`, save, and run Lighthouse again.

The bugs

There are 6 bugs. Find at least 4.

Hints, one per bug type:

- **Language:** what language should a screen reader speak?
- **The tab label:** look at the browser tab. Does it say what this page is?
- **Pictures:** a meaningful picture needs words. A decoration needs *empty* words.
- **Links:** read only the link texts, one after another. Do they make sense alone?
- **Forms:** every input needs a label you can click.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

The parcel label read aloud. A screen reader reads the label (`lang`, `title`, alt, link text) to someone who cannot see the parcel. If the label is wrong or empty, they are lost.

Done looks like

Lighthouse Accessibility is 100 (or close), the tab shows a real title, every link says where it goes, and clicking "Phone number" puts the cursor in the phone box.

Scoreboard

```
I found __ / 6. The reveal taught me: _________
```

Stuck?

1. Lighthouse is in F12. If you can't see the tab, click **>>** at the top of DevTools.

2. Open `notes/02-accessibility-and-meta.md`: look at "alt: meaningful vs decorative".
3. Ask the AI tutor: "What alt text should a decorative image have, and why?"

Fast?

Try **D2-FF5** Lost & Found.

D2-04 · Broken media

File: classwork/D2-04-broken-media/README.md

Time: 8 min · **Type:** Bug-hunt

Goal: Mend audio, video and iframe tags so that the radio, the TV and the window all work.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

1. Look at each section of the page. What should be there? What is missing?
2. Press play on every player that you can see.
3. Compare each `src` / `poster` with the real file names in the `media/` and `images/` folders.
4. Mend each problem in `index.html`. Save and test again.

The bugs

There are 5 bugs. Find at least 3.

Hints, one per bug type:

- **Buttons:** a player without buttons is invisible or cannot be started.
- **Spelling of file names:** dash - and underscore _ are different characters.
- **Folder names:** `image` is not `images`.
- **Extensions:** `.htm` is not `.html`.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

Two audio players that both play the bell, a video with a momo poster that plays, and a map inside the window.

Scoreboard

```
I found __ / 5. The reveal taught me: _________
```

Stuck?

1. Open F12 → **Console**. A red 404 line shows the exact path the browser could not find.
2. Open the `media/` and `images/` folders in VS Code and compare each name letter by letter.
3. Ask the AI tutor: "What does the controls attribute do on a video tag?"

Fast?

Go on to **D2-05**. Done with that too? Try **D2-FF5**.

D2-05 · Media corner

File: classwork/D2-05-media-corner/README.md

Time: 3 min · **Type:** 🔨 Build

Goal: Add a window into another room: an `<iframe>` that shows the Kathmandu map page.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

- ☐ In the section "A window: how to find us", replace the line `<p>Your iframe goes here.</p>`.
- ☐ Add an `<iframe>` with `src="kathmandu-map.html"`.
- ☐ Give it a `title` that says what is inside (a screen reader reads it).
- ☐ Give it `width="620"` and `height="520"`.
- ☐ **Bonus:** inside `<video>`, after `<source …>`, add fallback text: `Your browser cannot play this video.`

TV, radio, and a window into another room. `<video>` = the TV, `<audio>` = the radio, `<iframe>` = a window that shows another page.

Done looks like

The map page shows inside a box on your page. You can scroll inside the box.

Stuck?

1. Type `iframe` and press Tab: Emmet writes `<iframe src="" frameborder="0"></iframe>`. Delete `frameborder="0"` (it is old HTML).
2. The window shows "Cannot GET"? Check the spelling of `kathmandu-map.html`.
3. The title does not show on the page. That is normal: it is for screen readers. Check it in F12 → Elements.

Fast?

Try **D2-FF5** Lost & Found.

D2-06 · Stylesheet not loading

File: classwork/D2-06-stylesheet-not-loading/README.md

Time: 9 min · **Type:** Bug-hunt

Goal: Mend the link to the stylesheet and the CSS mistakes, so Didi's page wears its outfit.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

1. Look at the page. It should have a cream background, a big cherry title, a yellow marker on one sentence and a navy footer.
2. Mend the `<link>` in `index.html` first. Save. Now the outfit loads.
3. Then read `css/style.css` line by line and mend each mistake. Save after each one and look.

The bugs

There are 6 bugs. Find at least 4.

> **Fix this first:** bugs 1 and 2 (the `<link>` line). Until the stylesheet loads, you cannot see any of the others.

Hints, one per bug type:

- **The link:** check every letter of `rel` and every letter of the file name.
- **Braces:** every `{` needs its `}`.
- **Semicolons:** every declaration ends with `;`.
- **Spelling:** CSS uses American English.
- **Comments:** CSS has only one comment style: `/* … */`.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

Cream page, cherry h1, maroon h2 with a line under it, navy body text with space between lines, a yellow marker on "She still folds every momo by hand", and a navy footer with white text.

Scoreboard

```
I found __ / 6. The reveal taught me: _________
```

Stuck?

1. F12 → **Console**: a red 404 means the file name in `href` is wrong.
2. F12 → Elements → click an element → **Styles**. A crossed-out line with ⚠ is a declaration the browser dropped.

3. Ask the AI tutor: "Why would a CSS rule after a missing brace stop working?"

Fast?

Try **D2-FF4** Haunted stylesheet (10 CSS mistakes).

D2-07 · Restyle the bio

File: classwork/D2-07-restyle-the-bio/README.md

Time: 4 min · **Type:** Modify

Goal: Change 4 values in a stylesheet and watch Sunita's page change. Use the school-uniform file: `css/style.css`.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

- ☐ **Body background:** in `body`, change `white` to a soft colour (try `#fdf6ec`).
- ☐ **h1 colour:** in `h1`, change `black` to a strong colour (try `#990011`).
- ☐ **Paragraph size:** in `p`, change `16px` to `18px`.
- ☐ **Link colour:** in `a`, change `blue` to a dark colour (try `#14213d`).
- ☐ Save after **each** change and look at the page.
- ☐ **DevTools bonus:** F12 → Elements → click `<h1>` → in **Styles**, click the colour value and type a new one. The page changes, but the file does not. Copy the value you like into `style.css`.

The school uniform. One external stylesheet dresses every page that links to it.

Done looks like

Four changes are visible, and `index.html` has not changed at all. Show your neighbour your colours.

Stuck?

1. Nothing changes? Did you save `style.css`? Is the `<link>` in `index.html` still there?
2. A change does not work? Look for the `;` at the end of the line and the `}` at the end of the rule.
3. Hex colours start with `#` and have 6 characters: 0–9 and a–f.

Fast?

Change the font too: `font-family: Georgia, serif;`. Then try **D2-FF4**.

D2-08 · Selector mix-ups

File: classwork/D2-08-selector-mixups/README.md

Time: 8 min · **Type:** Bug-hunt

Goal: Make every selector find its element.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

1. Look at the page: which parts look plain when they should be styled?
2. For each plain part, find its rule in `css/style.css` and its tag in `index.html`. Do they match exactly?
3. Mend each mistake (in the CSS or the HTML). Save and check.

The bugs

There are 5 bugs. Find at least 3.

Hints, one per bug type:

- **Capital letters:** `card` and `Card` are different names.
- **Ids:** the name after `#` must be the same as `id="…"`.
- **Classes need a dot:** `.price` finds `class="price"`.
- **Commas:** a selector list cannot end with a comma.
- **Hover:** move the mouse over every link. Check the spelling of the part after `:`.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

A navy nav bar with white links, cherry Georgia headings, three white cards with red borders, big bold red prices, and "Order" links that turn red-and-white on hover.

Scoreboard

```
I found __ / 5. The reveal taught me: ________
```

Stuck?

1. F12 → Elements → click the plain element. If its rule is **missing** from the Styles panel, the selector does not match.
2. Read the "Calling students in class" slide: element, `.class`, `#id`.
3. Ask the AI tutor: "Why does a trailing comma in a selector list break the rule?"

Fast?

Go on to **D2-09**. Then try **D2-FF3** Specificity duel.

D2-09 · Selector target practice

File: classwork/D2-09-selector-target-practice/README.md

Time: 5 min · **Type:** 🔨 Build

Goal: Hit 5 targets, with one CSS rule each. Do not touch the HTML.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

- ☐ **Target 1:** every paragraph → grey text `#666666`. *(element selector)*
- ☐ **Target 2:** only the veg dishes → light green background `#dff3dc`. *(class selector)*
- ☐ **Target 3:** today's special → bold and `1.5rem`. *(id selector)*
- ☐ **Target 4:** only the links **inside the nav** → no underline. The "Back to top" link keeps its underline. *(descendant selector)*
- ☐ **Target 5:** only dishes that are **both** `dish` and `spicy` → `3px solid red` border. *(two classes, no space)*
- ☐ Write each rule under its comment in `css/style.css`. Save after each one.

Calling students in class: `p` = "everyone", `.veg` = "the green team", `#todays-special` = "Aarav, by name", `nav a` = "the green team in row 2".

Done looks like

Grey paragraphs; Veg chowmein and Aloo achar are light green; today's special is big and bold; the nav links have no underline but "Back to top" does; Aloo achar and Buff sekuwa have red borders.

Stuck?

1. Read `index.html` first: which `class` and `id` does each element have?
2. `nav a` (with a space) means "an `a` somewhere inside a `nav`". `.dish.spicy` (no space) means "one element with both classes".
3. Target hit the wrong things? F12 → Elements → click the element and read which rule matched.

Fast?

Try **D2-FF3** Specificity duel: predict who wins before you look.

D2-10 · Units & colours

File: classwork/D2-10-units-and-colours/README.md

Time: 8 min · **Type:** 🐞 Bug-hunt

Goal: Mend values: units, colours, font fallbacks.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

1. Look at the page. Read the "Done looks like" below and compare.
2. Read each value in `css/style.css`. Is it a value the browser understands?
3. Mend each mistake. Save and check.

The bugs

There are 5 bugs. Find at least 3.

Hints, one per bug type:

- **Units:** a size like `300` needs a unit (`px`, `rem`, `%`).
- **Hex colours:** only 0–9 and a–f.
- **Spaces:** a number and its unit stick together.
- **Fonts:** always give a fallback list that ends with a generic family.
- **Line height:** `1.5px` and `1.5` are very different.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

A white price board 300 px wide with a navy border, red prices, a bigger cherry special, an Arial font, and a note whose lines have space between them.

Scoreboard

```
I found __ / 5. The reveal taught me: ________
```

Stuck?

1. F12 → Elements → Styles: ⚠ and a crossed-out line = a value the browser could not read.
2. Look at the "Measuring tape" slide: px, %, rem, em.
3. Ask the AI tutor: "What is the difference between line-height 1.5 and 1.5px?"

Fast?

Go on to **D2-11**. Then try **D2-FF4**.

D2-11 · Theme the canteen menu

File: classwork/D2-11-theme-the-canteen-menu/README.md

Time: 5 min · **Type:** Modify

Goal: Dress the canteen menu in the TSI palette, with a safe font stack, rem sizes and readable line height.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

- ☐ **Palette:** cream `#FDF6EC` page background, navy `#14213D` text, cherry `#990011` for h1 and h2.
- ☐ **Font stack:** replace `"Times New Roman"` with `"Segoe UI", Tahoma, Arial, sans-serif`.
- ☐ **rem sizes:** body `16px` → `1rem`, h1 `40px` → `2.5rem`, h2 `24px` → `1.5rem`, `.note` `14px` → `0.875rem`.
- ☐ **Line height:** add `line-height: 1.6;` to `body`.
- ☐ **Bonus:** give the table header (`th`) a navy background and cream text.

Paint shop (colours) and **measuring tape** (units): pick from the shop, measure with the tape.

Done looks like

A cream page with navy text, cherry headings, a sans-serif font, comfortable line spacing, and no `px` font sizes left.

Stuck?

1. rem = "times the root size". The root is 16px, so `1.5rem` = 24px.
2. A colour does not show? Check the `#` and the 6 characters.
3. Font still looks like Times? Each font name with a space needs quotes, and names are separated by commas.

Fast?

Try **D2-FF4** Haunted stylesheet.

D2-12 · Broken cards

File: classwork/D2-12-broken-cards/README.md

Time: 8 min · **Type:** Bug-hunt

Goal: Mend box-model mistakes: size arithmetic, border shorthand, centring, round images.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save. Make the browser window wide (maximise it).

Task

1. Look at the page: three cards should sit side by side in one row, in a cream box in the middle of a grey page.
2. F12 → Elements → hover each box. Read the box diagram (content, padding, border, margin).
3. Mend each mistake in `css/style.css`. Save and check.

The bugs

There are 5 bugs. Find at least 3.

Hints, one per bug type:

- **Arithmetic:** add up width + padding + border for one card. Do three fit in the row?
- **Centring:** `margin: 0 auto` only works when the box has a width.
- **Padding:** some values are impossible.
- **Border shorthand:** width, **style**, colour. Is one missing?
- **Circles:** `border-radius: 50%` makes a circle only on a square.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Boxes being shipped: content = the item, padding = bubble wrap, border = the cardboard, margin = the gap between parcels.

Done looks like

Three cards in one row, a cream box centred on a grey page, an orange border around the note, red tags with space around the text, and round avatars.

Scoreboard

```
I found __ / 5. The reveal taught me: _________
```

Stuck?

1. F12 → Elements → **Computed** tab: the box diagram shows each number.
2. Read the "width trap: content-box vs border-box" slide.
3. Ask the AI tutor: "Why does border-radius 50% make an oval on my image?"

Fast?

Go on to **D2-13**. Then try **D2-FF1** TSI ID card.

D2-13 · Profile card

File: classwork/D2-13-profile-card/README.md

Time: 7 min · **Type:** 🔨 Build

Goal: Build a centred profile card for Sunita with the box model. Then show it on the projector.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

- ☐ `.card`: `width: 320px;` and `margin: 0 auto;` (centred)
- ☐ `.card`: `padding` (try `24px`) and a `border` (width, style, colour)
- ☐ `.card`: `border-radius` (try `16px`) and a white `background-color`
- ☐ `.card`: `box-sizing: border-box;` (the card stays exactly 320 px wide)
- ☐ `.avatar`: equal `width` and `height` (120 px) and `border-radius: 50%;` (round)
- ☐ `body`: a soft background colour, so the white card stands out
- ☐ **Bonus:** a coloured ring around the avatar, `text-align: center`, your own colours

Done looks like

A white card with rounded corners in the middle of the page, a round photo, and the text centred. In F12, the card measures exactly 320 px wide. **Show & tell:** 2–3 cards go on the projector.

Stuck?

1. Card stuck on the left? `margin: 0 auto` needs a `width`.
2. Photo is an oval? Width and height must be the same.
3. Card wider than 320 px in F12? You forgot `box-sizing: border-box`.

Fast?

Try **D2-FF1** TSI ID card: copy a design from a picture.

D2-14 · Nav won't behave

File: classwork/D2-14-nav-wont-behave/README.md

Time: 8 min · **Type:** Bug-hunt

Goal: Mend `display` mistakes so the Momo House header becomes a neat horizontal menu bar.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

Task

1. Look at the header. It should show the logo, then one row of 4 red menu buttons, and below that a cream hero with a centred "Order now" button.
2. Find the CSS rule for each part that looks wrong.
3. Mend each mistake in `css/style.css`. Save and check.

The bugs

There are 5 bugs. Find at least 3.

Hints, one per bug type:

- **Spelling of values:** read every `display` value letter by letter.
- **Missing things:** something is on the page in HTML but not on the screen. Who hid it?
- **Bullets:** a menu bar has no bullets.
- **The bench:** inline boxes can't be pushed up or down. Which value keeps them side by side *and* lets them have size?
- **Selectors:** does the class in the CSS exist in the HTML?

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

The logo at the top, 4 red buttons in one row with no bullets and no overlap, and a centred 200 px "Order now" button in the cream hero.

Scoreboard

```
I found __ / 5. The reveal taught me: _________
```

Stuck?

1. F12 → Elements → hover each link: the blue/green overlay shows its padding and margin.
2. Read the "queue vs the bench" slide.
3. Ask the AI tutor: "Why is margin-top ignored on my inline link?"

Fast?

Try **D2-FF2** Hover menu.

D2-15 · Style Momo House

File: classwork/D2-15-style-momo-house/README.md

Time: 28 min (5 checkpoints) · **Type:** 🔨 Build

Goal: Opening night! Dress Didi's restaurant with **one** stylesheet for all 4 pages: the school uniform.

Open `index.html` with Live Server (right-click → **Open with Live Server**). No Live Server? Double-click the file and press F5 after each save.

The 4 pages (`index.html`, `menu.html`, `order.html`, `about.html`) already have header, nav, main and footer, and the classes you need. You write **only** `css/style.css`. The instructor checks the room at each checkpoint.

Task

CP1 (about 0:06) · Link + base

- ☐ Add `<link rel="stylesheet" href="css/style.css">` inside `<head>` on **all 4 pages**.
- ☐ `body`: `margin: 0`, a background colour, a text colour, a font stack with a fallback, `line-height: 1.6`.
- ☐ `main`: `max-width: 960px; margin: 0 auto; padding: 16px;`
- ☐ Click through all 4 pages: they all change. That is the uniform.

CP2 (about 0:12) · Header + horizontal nav

- ☐ `.site-header`: a dark background and padding.
- ☐ `.menu`: `list-style: none; margin` and `padding` 0.
- ☐ `.menu li`: `display: inline-block;`
- ☐ `.menu a`: white, no underline, `display: inline-block`, padding.

CP3 (about 0:18) · Hero + h1

- ☐ `.hero-img`: `display: block; width: 100%; height: auto;`
- ☐ `.hero`: `text-align: center`, a background, padding at the bottom.
- ☐ `h1` and `h2`: your accent colour; `h1 2.5rem`.
- ☐ `.button`: an inline-block pill (background, white text, padding, `border-radius`).

CP4 (about 0:24) · Menu table + 3 dish cards (box model)

- ☐ `.price-table`: `border-collapse: collapse`; `th` and `td` get a border and padding; `th` gets a dark background.
- ☐ `.card`: `display: inline-block; box-sizing: border-box; width: 290px;` + padding, border, `border-radius`, margin.
- ☐ `.card img`: `width: 100%; height: auto;`
- ☐ **Bonus:** style the order form (`.order-form label` as block, inputs full width) and make the About photo round.

CP5 (about 0:28) · Footer + hover

- ☐ `.site-footer`: dark background, white centred text, padding.
- ☐ `:hover` on the nav links (and the button): change the background colour.
- ☐ Final check: click every nav link on every page. Everything wears the same uniform.

Done looks like

All 4 pages share one look: a navy header with the logo and a horizontal menu, a cream page, cherry headings, a full-width hero picture, a neat price table, 3 dish cards in one row, and a navy footer. Hovering the menu changes its colour. `css/style.css` is the **only** CSS file.

Stuck?

1. Only one page changed? The `<link>` must be on **every** page.
2. Menu still vertical? `display: inline-block` goes on `.menu li` (the items), not on `.menu`.
3. Third card drops to a new line? Add `box-sizing: border-box` to `.card`, or make the width a little smaller.

Fast?

Try **D2-FF2** Hover menu, or **D2-FF6** Bootstrap try-it (compare 100 lines of your CSS with Bootstrap's classes).

Fast finishers · Day 2

File: classwork/fast-finishers/README.md

Finished a classwork task early? Pick one of these. They are in **any order**. Each folder has a `README.md`. Solutions are in each `solution/` folder (the instructor shares them after class).

Id	Folder	Type	What you do	Good after
D2-FF1	`D2-FF1-tsi-id-card/`	Build	Copy Aarav's TSI ID card from a picture (`target.svg`): box model + fonts	D2-12, D2-13
D2-FF2	`D2-FF2-hover-menu/`	Modify	Make the Momo House menu change colour, background and underline on `:hover`	D2-14
D2-FF3	`D2-FF3-specificity-duel/`	Predict	8 puzzles: guess which CSS rule wins, then check in DevTools	D2-08, D2-09
D2-FF4	`D2-FF4-haunted-stylesheet/`	10 bugs	Chase 10 CSS mistakes out of the Kathmandu Ghost Walk page	D2-06, D2-10
D2-FF5	`D2-FF5-lost-and-found-bughunt/`	11 bugs	Semantic + a11y + CSS mistakes on the TSI Open Day page	D2-02, D2-03
D2-FF6	`D2-FF6-bootstrap-try-it/`	Modify	Add Bootstrap's `container`, `btn btn-primary` and `card` to a plain page	B10 demo

Rule: the One Rule still applies: you may only ship code you can explain, line by line.

D2-FF1 · TSI ID card

File: classwork/fast-finishers/D2-FF1-tsi-id-card/README.md

Time: fast finisher · **Type:** 🔨 Build

Goal: Recreate Aarav's TSI ID card from a picture, using the box model and fonts.

Open `index.html` with Live Server (right-click → **Open with Live Server**). Open `target.svg` in a second tab (or in VS Code) next to it.

Task

- ☐ Read the sizes written around the card in `target.svg`.
- ☐ `.id-card`: 360 × 220 px, `padding: 16px`, `border: 2px solid #14213D`, `border-radius: 12px`, white, centred. Use `box-sizing: border-box`.
- ☐ `.band`: navy background, white bold text, centred.
- ☐ `.avatar`: 80 px round, with a cherry `#990011` ring.
- ☐ `.details`: `display: inline-block` so it sits next to the photo. Text colour `#2A3F63`.
- ☐ `.name`: 26 px bold. `.stripe`: a cherry stripe at the bottom.
- ☐ You may **not** change `index.html`.

Done looks like

Put your page and `target.svg` side by side. A friend should not be able to tell which is which from 2 metres away.

Stuck?

1. Photo and text on top of each other? `.details` needs `display: inline-block`.
2. Card bigger than 360 px in F12? Add `box-sizing: border-box`.
3. Text not lined up with the photo? Try `vertical-align: middle` on both `.avatar` and `.details`.

Fast?

Try **D2-FF3** Specificity duel.

D2-FF2 · Hover menu

File: classwork/fast-finishers/D2-FF2-hover-menu/README.md

Time: fast finisher · **Type:** ✎ Modify

Goal: Make the Momo House menu answer the mouse: change colour, background and underline on `:hover`.

Open `index.html` with Live Server (right-click → **Open with Live Server**).

Task

- ☐ Add **one** new rule at the end of `css/style.css`: `.menu a:hover { … }`.
- ☐ On hover: a cherry background `#990011`.
- ☐ On hover: a different text colour (try `#ffcc66`).
- ☐ On hover: `text-decoration: underline;`.
- ☐ **Bonus:** change the colour of the bottom border on hover (`border-bottom-color`).

Done looks like

When the mouse is over a menu item, it changes colour and background and gets an underline. When the mouse leaves, it goes back.

Stuck?

1. Nothing happens? Check the colon: `a:hover`, no space before `hover`.
2. Only the text changes, not the box? The links need `display: inline-block` and padding (they already have it).
3. F12 → Elements → click a link → in Styles, click **:hov** and tick **:hover** to freeze the hover state.

Fast?

Try **D2-FF4** Haunted stylesheet.

D2-FF3 · Specificity duel

File: classwork/fast-finishers/D2-FF3-specificity-duel/README.md

Time: fast finisher · **Type:** 🔨 Predict + check

Goal: 8 puzzles: two CSS rules fight over one element. Predict the winner, then check in DevTools.

Open `css/style.css` first. **Do not open the page yet.**

Task

- ☐ For each puzzle, read the HTML element in `index.html` and its rules in `css/style.css`.
- ☐ Write your guess on paper: `1:` _____ `2:` _____ … `8:` _____.
- ☐ Now open `index.html` with Live Server. Count your correct guesses.
- ☐ For each wrong guess: F12 → Elements → click the element → Styles. Which rule is crossed out?
- ☐ Then read `solution/ANSWERS.md` (ask the instructor).

Done looks like

8 guesses on paper, checked against the page, and for each one you can say **why** it won.

Stuck?

1. Score each selector: count ids, then classes, then elements. Compare from the left.
2. Same score? The rule written later wins.
3. Puzzle 6 and 7 are special: one is about `style=""`, one is about inheritance.

Fast?

Try **D2-FF4** Haunted stylesheet.

D2-FF4 · Haunted stylesheet

File: classwork/fast-finishers/D2-FF4-haunted-stylesheet/README.md

Time: fast finisher · **Type:** Bug-hunt

Goal: Chase 10 ghosts out of a stylesheet. Every CSS mistake type from today is here.

Open `index.html` with Live Server (right-click → **Open with Live Server**).

Task

1. Look at the page, then read `css/style.css` from top to bottom.
2. Mend each mistake. Save after each one and look at the page.
3. All 10 mistakes are in `css/style.css`. Do not change `index.html`.

The bugs

There are 10 bugs. Find at least 8.

Hints, one per bug type:

- **Braces:** every `{` needs a `}`.
- **Units:** sizes need a unit, with no space before it.
- **Selectors:** capital letters, and ids that do not exist.
- **Spelling:** of properties and of values.
- **Shorthand:** a border needs a style.
- **Comments:** CSS has only `/* … */`.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

A purple nav bar, a big ghost-green title, a large italic subtitle, red-bordered story boxes, gold headings, a 140 px badge on its own line, a green rules list, a green marker on "no bell in the courtyard", and a grey centred footer.

Scoreboard

```
I found __ / 10. The reveal taught me: _________
```

Stuck?

1. F12 → Elements → Styles: crossed-out lines with ⚠ are dropped declarations.
2. A whole rule missing from Styles? Its selector does not match (or a brace/comment above it is broken).
3. Read `notes/09-debugging-css.md`: "what the browser drops".

Fast?

Try **D2-FF5** Lost & Found.

D2-FF5 · Lost & Found

File: classwork/fast-finishers/D2-FF5-lost-and-found-bughunt/README.md

Time: fast finisher · **Type:** Bug-hunt

Goal: Find 11 mistakes from every part of today: semantic tags, accessibility, media, CSS values and selectors.

Open `index.html` with Live Server (right-click → **Open with Live Server**).

Task

1. Check the page with your eyes, the **Tab** key and **Lighthouse** (F12 → Lighthouse → Accessibility).
2. Read `index.html` against the rules for semantic tags and headings.
3. Read `css/style.css` for values and selectors that do not work.
4. Mend each mistake. Save and check.

The bugs

There are 11 bugs. Find at least 9.

Hints, one per bug type:

- **Semantic:** is each box named correctly? Can a footer hold a header?
- **Meaning:** bold that only looks bold vs bold that means "important".
- **Headings:** h1 → h2 → h3, no jumps.
- **Controls:** can Tab reach the Apply button?
- **Tables:** header cells are `th`, not `td` + `b`.
- **Frames:** every iframe needs a title.
- **CSS values:** units, and `:` not `=`.
- **Selectors:** a space in a selector means "inside". A comma does not belong in `class="…"`.

How you can see a bug (the tag in the answer key):

Tag	Meaning	Tool that shows it
V	You can **see** it on the page	your eyes
D(HH)	HTMLHint underlines it in VS Code / Problems panel (Ctrl+Shift+M)	HTMLHint
D(F12)	Only the Elements panel shows what the browser actually built	Chrome DevTools
D(LH)	Lighthouse → Accessibility report finds it (works offline)	Chrome DevTools
D(W3C)	Only the W3C validator reports it	validator.w3.org (Wi-Fi bonus)
R	**Rule check**: no tool fires; read the code against the rule	your brain + the notes

Done looks like

A cream page with a bigger intro line, a real nav, h2 Programme, a fee table with a navy header row, a red pill Apply button you can Tab to, a white card with a red left edge, and a map with a title.

Scoreboard

```
I found __ / 11. The reveal taught me: ________
```

Stuck?

1. Lighthouse finds 2 of them. Tab finds 1. Your eyes find the CSS ones.
2. F12 → Elements: click the card. Which classes does the browser see?
3. Read `notes/01-semantic-html5.md` and `notes/09-debugging-css.md`.

Fast?

Try **D2-FF6** Bootstrap try-it.

D2-FF6 · Bootstrap try-it

File: classwork/fast-finishers/D2-FF6-bootstrap-try-it/README.md

Time: fast finisher · **Type:** ✎ Modify

Goal: Use ready-made parts (Bootstrap) instead of raw wood (your own CSS). Compare the two.

Open `index.html` with Live Server (right-click → **Open with Live Server**). The Bootstrap file is already in `css/` (no internet needed).

Task

- ☐ In `<head>`, add `<link rel="stylesheet" href="css/bootstrap.min.css">`. Save. The font changes already.
- ☐ Wrap everything in `<body>` in `<div class="container py-4">` … `</div>`.
- ☐ Give the first link `class="btn btn-primary"`.
- ☐ Turn the `<div>` with "This week" into a card: `<div class="card mt-4">` with an inner `<div class="card-body">`. Add `class="card-title"` to its h2 and `class="card-text"` to its p.
- ☐ **Bonus:** `class="lead"` on the first paragraph; `btn btn-outline-primary` on "I'm coming".
- ☐ **Compare:** how many lines of CSS did you write? How many did your D2-13 card need?

IKEA for websites. Bootstrap gives you ready-made parts. Fast, but every site looks a bit the same, and you still need to understand CSS to change them.

Done looks like

A centred page with a blue button and a neat card, and you wrote **zero** lines of CSS.

Stuck?

1. Nothing changes? Check the `<link>` path: `css/bootstrap.min.css`.
2. Class names are separated by **spaces**: `btn btn-primary`.
3. Want more parts? With Wi-Fi, look at getbootstrap.com → Docs → Components.

Fast?

You finished everything! Help a neighbour, or start **D2-HW2** (style your portfolio).